Sangam Kumar Gupt

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Summary

Generative AI / ML Engineer — B.Tech CSE (Data Science), 2026 graduate; skilled in building and deploying RAG pipelines, CNN-based ML systems, and LLM-integrated applications end-to-end.

Technical Skills

- **Languages:** Python, SQL
- **AI/ML:** TensorFlow, Keras, Scikit-learn, Transfer Learning, NLP, Computer Vision, OpenCV
- **Generative AI & LLMs:** RAG Pipelines, Prompt Engineering, LLM Integration, Embeddings, Hugging Face, Google Gemini API, Groq API
- **Vector Databases:** FAISS, ChromaDB
- **Frameworks & APIs:** LangChain, FastAPI, Flask, Streamlit, REST APIs
- **Data & Tools:** Pandas, NumPy, Jupyter Notebook
- **DevOps & Platforms:** Git, GitHub, Docker (Basic), Vercel
- **Other:** Chrome Extension APIs, Model Deployment, Agile

Internship Experience

Artificial Intelligence & Machine Learning Engineer Intern Jan 2026 – Mar 2026

SmartBridge

- Developed and implemented a data preprocessing pipeline that reduced model training time by 40% through optimized feature selection and cleaning techniques.
- Deployed PlantCareAI as the capstone project, applying Transfer Learning, CNN inference, and LLM-based treatment generation using Google Gemini Pro.
- Designed end-to-end ML pipelines covering image preprocessing, model inference, and API integration using Flask and TensorFlow.

Projects

PlantCareAI – AI Plant Disease Detection & Assistant Jan 2026 – Mar 2026 (GitHub)

Python, Flask, TensorFlow, Transfer Learning, Google Gemini Pro, REST API

- Engineered a full-stack ML web app using Transfer Learning (TensorFlow/Keras CNN) to detect **40+** plant diseases across 10 crops with **95%+** accuracy and sub-1-second inference time.
- Architected an end-to-end inference pipeline: image upload → preprocessing (224×224px) → CNN prediction → disease-specific treatment generation via LLM.
- Integrated **Google Gemini Pro** with session-based context for personalized agricultural Q&A — eliminated repeated user input via context persistence.
- Delivered mobile-ready Bootstrap UI with drag-and-drop upload, diagnosis history, and chatbot panel; deployed on Vercel.

SummarAIze – Smart Assistant for Research Documents Oct 2024 (GitHub)

LangChain, Streamlit, Groq API, FastAPI, FAISS, Embeddings, Python

- Created an AI-powered document assistant supporting **50+** page PDFs and TXT files using a Generative AI and RAG architecture.
- Implemented a RAG pipeline with **FAISS** vector database and sentence embeddings for sub-second semantic retrieval across **100k+** token corpora.
- Constructed a modular **FastAPI** backend with document chunking, embedding generation, and context-aware LLM response via Groq API; reduced API latency to under 2 seconds.
- Crafted a Streamlit frontend delivering instant summaries, source-cited Q&A, and auto-generated quizzes — improving document review efficiency by **60%+**.

Talk2Tube – YouTube Q&A Assistant (RAG Based) Nov 2024 (GitHub)

Python, LangChain, ChromaDB, FastAPI, JavaScript, Chrome Extension APIs

- Built a Chrome Extension enabling natural language Q&A on YouTube videos using a RAG pipeline with LLM-powered context-aware responses.
- Leveraged **LangChain** and **ChromaDB** for real-time transcript indexing, semantic search, and response generation across videos up to 3 hours.
- Achieved timestamped retrieval accuracy of **90%+** on benchmark Q&A tests by optimizing chunk size and embedding overlap parameters.
- Established an interactive side-panel UI with chat history persistence; embedded **10k+** token transcripts in under 5 seconds on average.

Education

Noida Institute of Engineering and Technology, Greater Noida, UP Nov 2022 – 2026

Bachelor of Technology in Computer Science (*Data Science*) – CGPA: 7.25

C R T S V M I C S K S Bazar, Maharajganj, UP May 2018 – Jun 2020

Intermediate

Certifications

Data Analysis with Pandas and Python – Infosys Springboard Oct 2024

Introduction to Large Language Models – IBM SkillsBuild Feb 2026